NIKITA RAJANEESH

(765) · 838 · 9720 ⋄ nr2937@columbia.edu ⋄ nrajanee.github.io

Machine learning researcher and engineer with experience in agent and model evaluation, multimodal models, test-time adaptation, and ML infrastructure (4+ years in industry and 3+ years in academia).

EDUCATION

Columbia University (New York, NY) *Sept 2023 - May 2025*
Advanced Master's Research program (advised by Prof. Richard Zemel)
GPA: 3.91/4.0
Selected Coursework: Deep learning, Computational Learning theory, Continual learning, Datasets in ML, Computational aspects of Robotics

Purdue University (West Lafayette, IN) *Aug 2016 - May 2020*
BS in Computer Science & Minor in Mathematics
GPA: 3.82/4.0
Selected Coursework: Randomized Algorithms (*Graduate-level*), Natural Language Processing (*Graduate-level*), Machine Learning, Artificial Intelligence, Analysis of Algorithms, Compilers, Linear Algebra

PUBLICATIONS

Test-Time Warmup for Multimodal Large Language Models. **Nikita Rajaneesh**, Thomas P Zollo, Richard Zemel. https://arxiv.org/abs/2509.10641

Towards effective discrimination testing for generative AI. FAccT, 2025. Thomas P Zollo, **Nikita Rajaneesh**, Richard Zemel, Talia B. Gillis, and Emily Black. https://arxiv.org/abs/2412.21052.

EXPERIENCE

Machine Learning Research Engineer - Arklex.AI July 2025 - Present
New York, NY

- Own the end-to-end agent evaluation component of flagship product that covers scenario design, conversation simulation, and automated agent evaluation and turns ad-hoc agent testing into a systematic, repeatable evaluation framework.
- Built a taxonomy of agent-behavior failure modes plus tooling to surface unique, high-signal errors and trace each back to targeted code-level fixes, tightening the debug loop for agent building.
- Developed a novel NLP framework to adaptively generate scenarios from real conversations and increased coverage of real errors by 15% for a leading education company.
- Implemented an ML algorithm that improves alignment of LLM judges to domain experts by 20%.

Machine Learning Researcher - Columbia University Sept 2023 - May 2025
New York, NY

- Developed an unsupervised test-time adaptation method for Multi-modal LLMs (MLLMs) that finetunes on a weakly supervised auxiliary task to boost generalization in label-scarce domains like medicine and yielding relative gains of 4.03% on MMMU, 5.28% on VQA-Rad (medical), and 1.63% on GQA.
- Demonstrated how variability in bias-testing methods (e.g., red-teaming) can let unfair models appear compliant, and how single-turn evaluations miss biases emerging in multi-turn interactions.
- Benchmarked state-of-the-art LLM uncertainty-quantification techniques (semantic uncertainty, eigenscore) in the multi-modal setting, extending their evaluation beyond text-only models.
- Designed a computationally efficient UQ method leveraging image-informed priors that matches existing LLM-based UQ baselines while cutting computational overhead.
- Built an algorithm to reconstruct the weights of black-box CNNs.

Software Engineer, AI/ML - Determined AI (HPE company) Feb 2022 - June 2023
Chicago, IL

- Developed software to enable users to customize hyperparameter-tuning with Determined's Deep Learning platform. Designed the software to ensure fault tolerance and distributed computation.
- Developed framework for an adversarial library toolkit that is integrable with deep learning platform.
- Wrote Python SDK and Go API for user management and authentication for model registry.
- Built functionality to delete checkpoints saved during model training.
- Built a tool to enable easy debugging of trials in model experiments.

Software Engineer - Morningstar, Inc. Aug 2020 - Feb 2022
Chicago, IL

- Developed software (using vaderSentiment and spaCy) to perform Sentiment Analysis on fund reviews.
- Developed an audit process (with AWS architecture) which collects metadata of tables in the Datalake.
- Worked with AWS lambda, AWS Glue jobs and Spark to parse and write AWS s3 access and cloudtrail logs to parquet files.

Machine Learning Researcher - Purdue University Jan 2020 - Aug 2020
West Lafayette, IN

- Extended the Hardt–Price–Srebro equalized-odds framework to the special case of scoring systems with non-decreasing conditional event probabilities, proving a universally optimal fair classifier exists in this setting.
- Showed this optimal classifier reduces to a single randomized threshold per demographic group, making it simple, explainable to policymakers, and computable in polynomial time.
- Co-authored the resulting paper, *On Equalized Odds in Supervised Learning, for the Special Case of Non-Decreasing Conditional Event Probabilities*, with Prof. Kent Quanrud.

Manuscript: https://nrajanee.github.io/files/equalizedodds_ced.pdf.

Software Engineering Intern - Morningstar, Inc. June 2019 – Aug 2019
Chicago, IL

- Developed software that will allow users to do analytics on the usage data of the Datalake.
- Developed software that will help users get access to the glue catalog in Datalake by using AWS Glue API and Apache Airflow.

Software Engineering Intern - Jobcase, Inc. June 2018 - Aug 2018
Boston, MA

- Developed a "view history" functionality using Java Hibernate in an AngularJS webapp called "Scheduler" to allow a user to record changes to a scheduled process.
- Developed a regular expressions based approach to automatically populate job requirements' fields to reduce job search time for a user. Used ElasticSearch and developed a parsing tool in Java to test and analyze the proposed approach.

SKILLS

Languages	Python, Go, C/C++, Java, TypeScript, Scala, SQL, R, MATLAB
Tools	PyTorch, Huggingface, vLLM, Tensorflow, AWS, Docker, Pandas, Spark, Numpy